

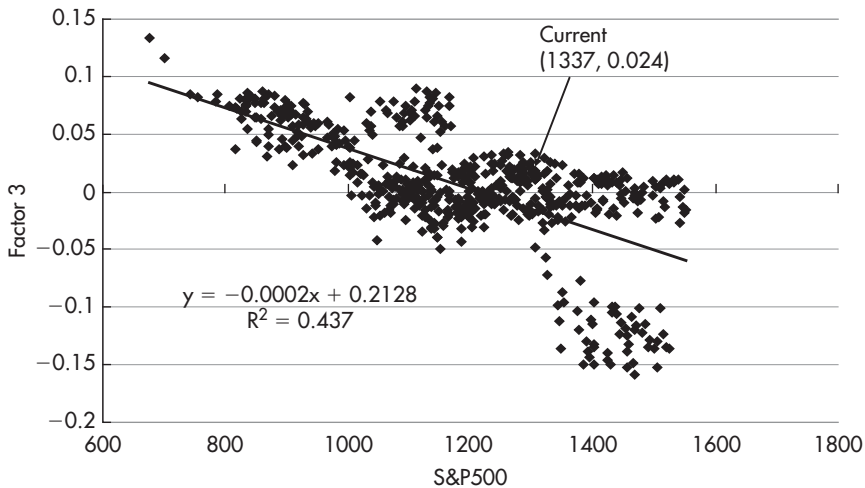


FIGURE 3.30 Regression of factor 3 of the PCA on currencies versus the S&P500 index.

Sources: data – ECB, Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 25 Jul 2011, weekly data.

neutral combination of currencies can be expected to work, as it could be mainly a function of factor 4 rather than factor 3. In addition, we would like to improve our return by choosing a combination of currencies with a high two-factor residual. These residuals are shown in Figure 3.31, and thus both a JPY–CHF–SEK and a JPY–SGD–SEK PCA-neutral combination of currencies (similar to a butterfly on the yield curve) seem attractive. However, only the first one has a strong correlation to the third factor, so we choose this one.

Now, we can formulate our trading strategy, which is to trade a two-factor PCA-neutral portfolio of JPY, CHF, and SEK versus the S&P500 index. The hedge ratio between the portfolio of currencies and the S&P500 index is given by the slope of the regression line, while the weightings of the currencies in the portfolio are determined by the conditions of neutrality versus factors 1 and 2 of the PCA on currencies. The result is the exposure to the residual of a regression between the portfolio of currencies and the S&P500 index, whose time series is shown in Figure 3.32.

Of course, we can now run a mean reversion model on that series and judge whether the statistical properties, like speed of mean reversion, seem sufficiently good to enter the trade. Figure 3.33 depicts the actual performance of the trade versus the forecast of the OU model. Since the actual performance followed the lower 2-sigma band of the OU simulation closely, one can say that an investor entering the trade was quite lucky to be able to realize the

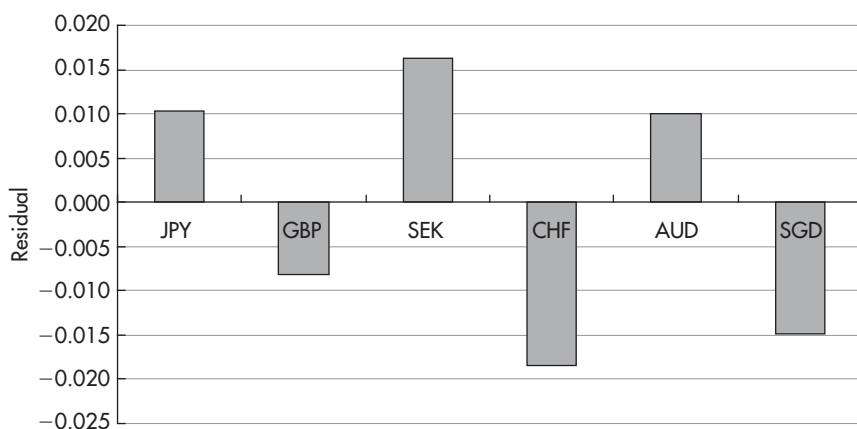


FIGURE 3.31 Two-factor residuals of a PCA on currencies.

Sources: data – ECB; chart – Authors.

Data period: 4 Jan 1999 to 25 Jul 2011, weekly data; current residuals as of 25 Jul 2011.



FIGURE 3.32 Residual of a regression between a PCA-neutral portfolio of currencies and the S&P500 index.

Sources: data – ECB, Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 25 Jul 2011, weekly data.

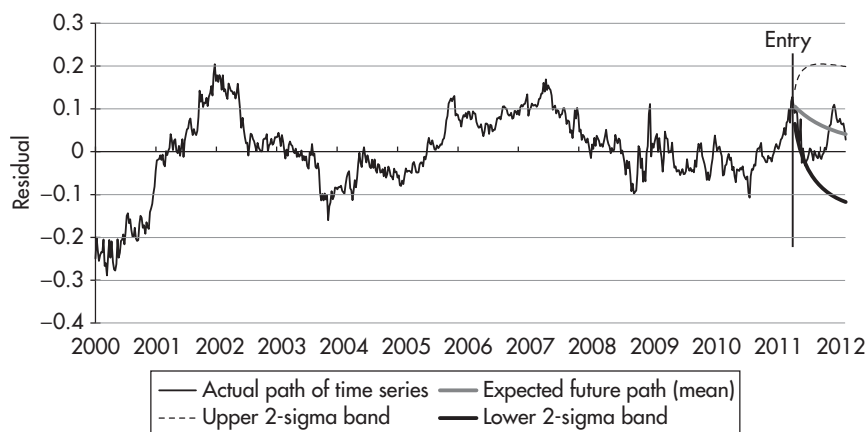


FIGURE 3.33 Actual performance of the trade versus the OU model forecast.

Sources: data – ECB, Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 25 Jul 2011, weekly data as input in OU model; forecast period as output of OU model versus weekly market data from 1 Aug 2011 to 4 Jun 2012.

profit potential from the time series returning to its mean (i.e. the residual of the regression disappearing) soon.

This example demonstrates how PCA both yields new insights into a variety of markets and is the basis for new trading opportunities. By producing time series (like factor 3) that are not directly observable in the market but can be traded nevertheless (in this example through a two-factor neutral portfolio), PCA opens the door to a multitude of new and creative analyses and trading opportunities. The advantages of these new positions, besides their analytical challenges and illuminating results, are that they offer:

- a good chance to find a source of profit which is uncorrelated to more common positions;
- trading ideas that are unlikely to be analyzed by many people and therefore likely to be profitable. While lots of analytical effort is invested in 2Y-5Y-10Y butterflies on the yield curve, probably no one ever thought of the trade developed above. Hence, PCA is also a tool to find value off the beaten track, as it allows constructing new trade ideas individually.

Our goal is not to promote trading stock indices versus portfolios of currencies but to encourage using PCA as a compass to venture into unknown territory, where golden fruits may be growing on the trees.

Multivariate Mean Reversion

INTRODUCTION

In the previous two chapters, we discussed ways to model univariate series that exhibit mean reversion and ways to model multivariate series that exhibit correlation. In this chapter, we discuss ways to model multivariate series that exhibit both mean reversion and correlation. More specifically, we discuss the *multivariate Ornstein–Uhlenbeck* (MVOU) process.

The MVOU process is capable of capturing a richer set of behaviors than are the univariate Ornstein–Uhlenbeck (OU) process and principal components analysis (PCA), even when the OU process and PCA are used together. For example, as we’ll see below, the MVOU process is capable of modeling nonmonotonic behavior in the path of expected values over time, whereas this behavior is not a feature of the OU process or of PCA.

Another important characteristic of the MVOU process is that it can model horizon-dependent correlation, i.e. correlation that depends on the elapsed time between observations. As we’ll see, it’s not unusual for market data to exhibit horizon-dependent correlations, a characteristic that neither the univariate OU process nor PCA can model, even when used together.

The cost of this additional capability is that the MVOU process is more complicated than either the OU process or PCA, and it’s more difficult to implement in practice. However, this additional difficulty is manageable. For example, all the calculations displayed in this chapter were performed in an Excel spreadsheet, including the multivariate optimization. And in our experience, the additional capabilities of the MVOU model are often well worth the additional effort.

In this chapter, we’ll first discuss the mathematics and intuition behind the MVOU process. Then we’ll consider some practical examples featuring two applications of the MVOU process to fixed income markets.

Mathematics

The mathematics of the MVOU process are similar in many respects to those of the OU process. But there are some important differences due to the

multivariate nature of the MVOU process. In addition, some of the quantities that must be calculated to implement the MVOU process can be a bit tricky, and we'll discuss these in separate text boxes.¹

Stochastic Differential Equation

For a set of m variables given by the vector, $\mathbf{X}_t$, the stochastic differential equation (SDE) for the MVOU process can be written:

$$d\mathbf{X}_t = -\boldsymbol{\Theta}(\mathbf{X}_t - \boldsymbol{\mu})dt + \mathbf{S}d\mathbf{B}_t$$

where

$-\boldsymbol{\Theta}$ is an $m \times m$ matrix, which we'll refer to as the *transition matrix*

$\boldsymbol{\mu}$ is an m -element vector containing the means of the m variables
 dt denotes an instantaneous change in time

$\mathbf{S}$ is an $m \times m$ matrix, which we'll refer to as the *scatter matrix*

$d\mathbf{B}_t$ is the instantaneous change in an m -element vector of independent Brownian motions.

It's instructive to consider this equation in scalar notation as well as in matrix notation. In, particular, let's consider an example with three variables, index by $i = 1, 2, 3$, such that X_{it} and B_{it} represent the i -th elements of $\mathbf{X}_t$ and $\mathbf{B}_t$. Denote the elements of $\boldsymbol{\Theta}$ and $\mathbf{S}$ by θ_{ij} and S_{ij} for $i, j = 1, 2, 3$. To simplify the expressions, and without loss of generality, we'll assume that the long-run means for each of the three variables are equal to zero – i.e. $\boldsymbol{\mu} = \mathbf{0}$. Then, in scalar notation, we have

$$dX_{1t} = (-\theta_{11}X_{1t} - \theta_{12}X_{2t} - \theta_{13}X_{3t})dt + S_{11}dB_{1t} + S_{12}dB_{2t} + S_{13}dB_{3t}$$

$$dX_{2t} = (-\theta_{21}X_{1t} - \theta_{22}X_{2t} - \theta_{23}X_{3t})dt + S_{21}dB_{1t} + S_{22}dB_{2t} + S_{23}dB_{3t}$$

$$dX_{3t} = (-\theta_{31}X_{1t} - \theta_{32}X_{2t} - \theta_{33}X_{3t})dt + S_{31}dB_{1t} + S_{32}dB_{2t} + S_{33}dB_{3t}$$

If θ_{11} is positive and $\theta_{12} = \theta_{13} = 0$, then X_{1t} follows a simple univariate Ornstein–Uhlenbeck process.² But let's assume that θ_{12} is positive. Then X_{1t} will have a tendency to decline when X_{2t} is high, and X_{1t} will have a tendency

¹In the first section of this chapter, we follow the excellent presentation provided in Meucci (2010).

²Since all the B_{it} terms are normal, their weighted sum is normal, and the univariate SDE for X_{1t} can be written in the form of a univariate OU process.

to increase when X_{2t} is low. In this case, X_{1t} will exhibit a tendency to move away from X_{2t} . We sometimes talk about X_{1t} being repulsed by X_{2t} . On the other hand, if θ_{12} is negative, then X_{1t} will have a tendency to increase when X_{2t} is high, and X_{1t} will have a tendency to decrease when X_{2t} is low. In this case, we sometimes refer to X_{1t} being attracted by X_{2t} .

If the value of θ_{21} is zero, then this effect is asymmetric, with X_{1t} ‘paying attention to’ X_{2t} but with X_{2t} ‘ignoring’ X_{1t} . But if θ_{21} is negative, then X_{2t} will be attracted to X_{1t} .

Any combination is conceivable in this case. One variable could be attracted to a second, while the second is repulsed by the first. Or the second variable could ignore the first. Typically, we’d expect any attraction between two variables to be mutual, but that need not be the case.

The elements of the scatter matrix $\mathbf{S}$, S_{ij} , impart correlation between the elements of $\mathbf{X}_t$. But they’re not the only way that the elements of $\mathbf{X}_t$ could be correlated. Even if all the off-diagonal elements of $\mathbf{S}$ were equal to zero, the elements of $\mathbf{X}_t$ could be correlated by virtue of the off-diagonal elements of Θ . In other words, even if the elements of $\mathbf{X}_t$ aren’t subject to similar shocks imparted by the Brownian motions, they will exhibit correlation if they are attracted or repulsed by one another.

It’s particularly interesting that off-diagonal elements of Θ impart horizon-dependent correlations between the elements of $\mathbf{X}_t$. In other words, the correlation between X_{1t} and X_{2t} could be negative over a horizon of one day, zero over a horizon of one month, and positive over a horizon of six months. We’ll see that this capability of the MVOU model is useful in capturing horizon-dependent correlations observed in actual market data.

The Differential Equation Within the MVOU Process

For intuition, let’s consider the drift coefficient, $-\Theta(\mathbf{X}_t - \boldsymbol{\mu})$, and the diffusion coefficient, $\mathbf{S}$, in turn.

Without the diffusion coefficient, we have the differential equation, $d\mathbf{X}_t = -\Theta(\mathbf{X}_t - \boldsymbol{\mu})dt$, with the solution $\mathbf{X}_v = \boldsymbol{\mu} + e^{-\Theta(v-t)}(\mathbf{X}_t - \boldsymbol{\mu})$, for $v > t$. In the event the mean vector, $\boldsymbol{\mu}$, were simply the $\mathbf{0}$ vector, this equation reduces to $\mathbf{X}_v = e^{-\Theta(v-t)}\mathbf{X}_t$.

Note that $e^{-\Theta(v-t)}$ is a matrix. In fact, for some fixed amount of time, $\tau = v - t$, we can define a matrix, $\mathbf{A} = e^{-\Theta\tau}$. Then we could approximate the value of $\mathbf{X}_{t+i\tau}$ by repeated application of the matrix, $\mathbf{A}$. That is, $\mathbf{X}_{t+i\tau} = \mathbf{A}^i\mathbf{X}_t$, for $i = 1, 2, 3 \dots$

All the elements of $\mathbf{X}_t$ will shrink toward zero over time (again, assuming $\boldsymbol{\mu} = \mathbf{0}$), provided the absolute values of the eigenvalues of $\mathbf{A}$ are all less than one. Given that $\mathbf{A} = e^{-\Theta\tau}$, this eigenvalue condition for $\mathbf{A}$ will be satisfied when all the eigenvalues of Θ are positive.

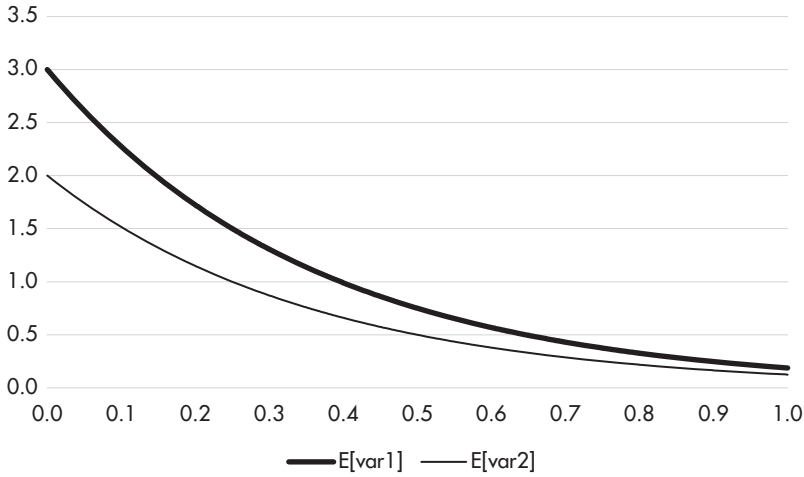


FIGURE 4.1 Expected values over time.

Source: Authors.

Let's consider this in a bivariate example. In particular, let's specify a transition matrix

$$\Theta = \begin{bmatrix} \ln(2)/0.25 & 0 \\ 0 & \ln(2)/0.25 \end{bmatrix}$$

so that each variable has a half-life of three months (0.25 years).³ Let's also assume that the starting values for the two variables are 3 and 2. Then the path of expected values given by the transition matrix, Θ , is shown in Figure 4.1.

The expected spread between the two variables (var1 – var2) is shown over time in Figure 4.2.

Because both variables have identical half-lives, the spread also follows a univariate Ornstein–Uhlenbeck process with a half-life of three months.

Now let's see what happens if we specify a half-life of one month for the second variable while leaving the half-life of the first variable at three months. In other words, we specify the transition matrix

$$\Theta = \begin{bmatrix} \ln(2)/0.25 & 0 \\ 0 & \ln(2) * 12 \end{bmatrix}$$

Figure 4.3 shows the path of the two expected values over time.

³As a reminder, the half-life is given by $\ln(2)/\theta$.

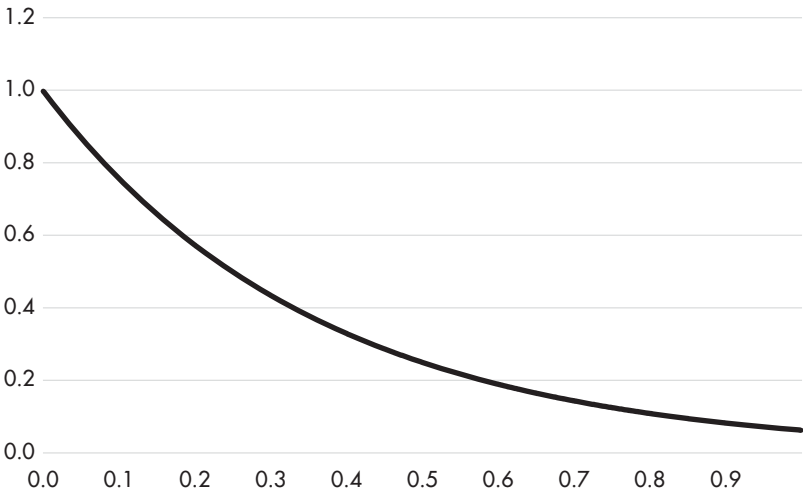


FIGURE 4.2 Expected spread over time.
Source: Authors.

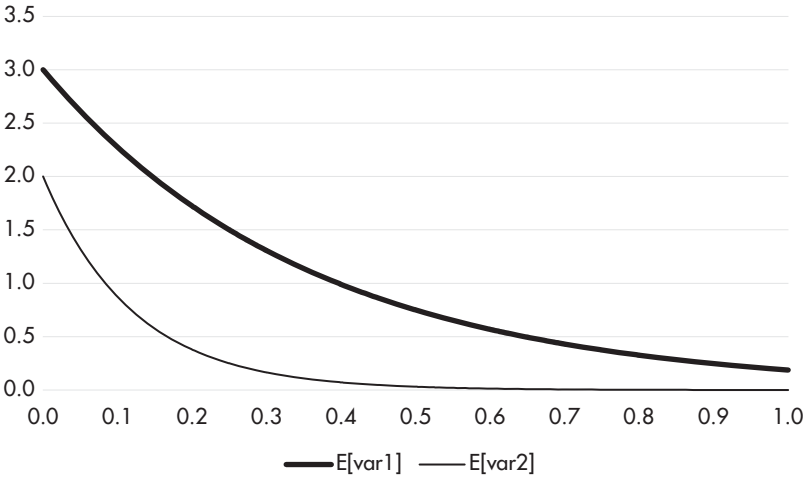


FIGURE 4.3 Expected values over time, given different half-lives.
Source: Authors.

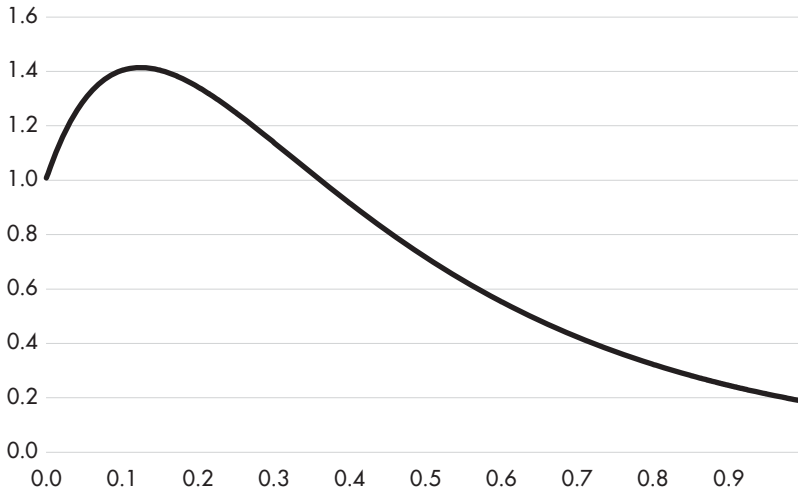


FIGURE 4.4 Expected spread over time, given different half-lives.

Source: Authors.

Figure 4.4 shows the path of the expected spread over time.

Since the spread is a linear function of two stationary variables, it's also stationary. However, it's clear from Figure 4.4 that the spread can't be written in the form of a univariate Ornstein–Uhlenbeck process. In particular, note that the path of expected values for this spread isn't even monotonic. It increases for some weeks before peaking and beginning a convergence to its long-run mean of zero.

Clearly, this behavior has very different implications for relative value analysis than if the spread could be modeled as a univariate OU process.

Now let's specify that the variable with the shorter half-life is attracted to the variable with the longer half-life. In particular, we'll specify as the transition matrix

$$\Theta = \begin{bmatrix} \ln(2)/0.25 & 0 \\ -8 & \ln(2) * 12 \end{bmatrix}$$

The path of expected values for the two variables is shown in Figure 4.5.

The path of expected values for the spread is shown in Figure 4.6.

Note that the path of expected values for the spread crosses the long-run mean of zero and then converges to this long-run mean from below. Intuitively, one might conclude that the spread has 'overcorrected' or 'overshot' its long-run mean. But while it's probably accurate to say that the spread is expected to overshoot its long-run mean, it's not useful to think of this as an

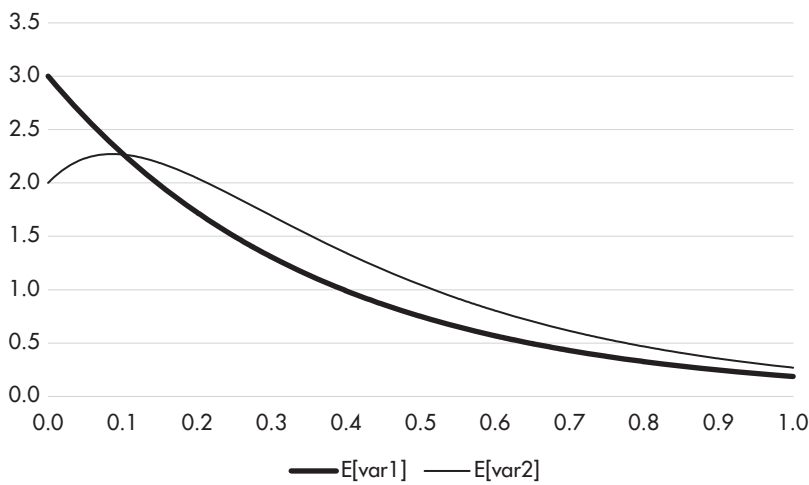


FIGURE 4.5 Expected values over time, given different half-lives and a one-way attraction
Source: Authors.

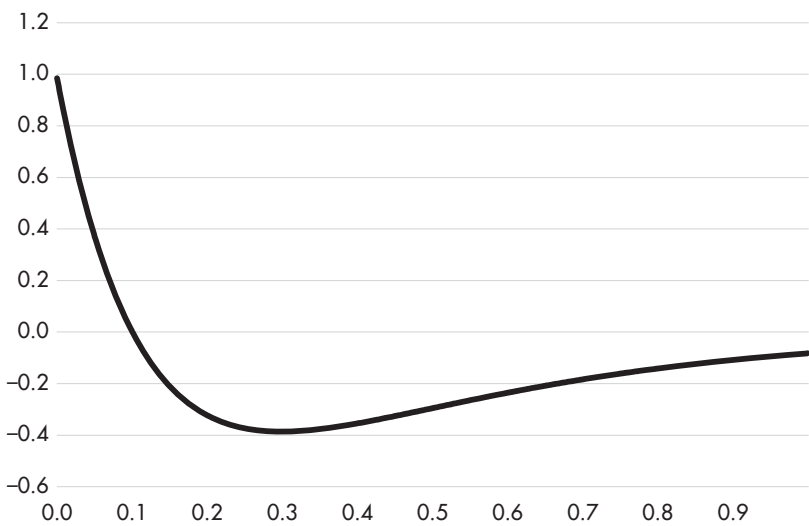


FIGURE 4.6 Expected spread over time, given different half-lives and a one-way attraction.
Source: Authors.